Corpus / Domain Resources

PDF Metallogenic Literature

Parsing and Cleaing

Traceable JSON Chunks

Taxonomy | Relation Schema | Domain Lexicon

TSC-NER

A. Progressive Boundary Extraction — Sec.4.1

BERT Encoder

Start Head

End Head

... arsenian pyrite ...

Unlabeled Chunks

Chunk-Level Scoring (Confidence / Uncertainty)

High-stability Pseudo Labels

Low-stability Samples

LLM Boundary Correction

Update BERT

Curriculum Self-Training

B.Hierarchical Typing — Sec.4.2

... arsenian pyrite ...

Context (Chunk)

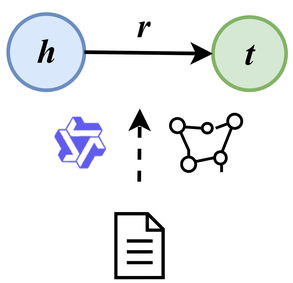
Entity Taxonomy

LLM Typing

Composition (Coarse) → Mineral (Fine)

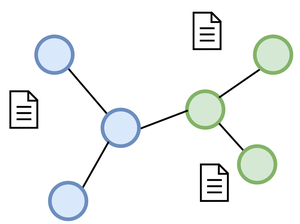
MAGKG Construction — Sec.4.3

Relation Extraction



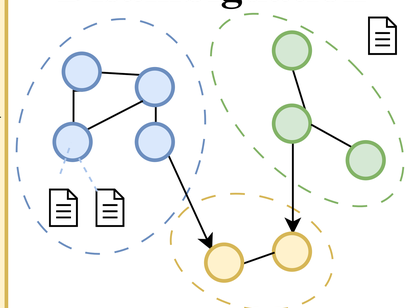
Schema Check

Evidence Graph



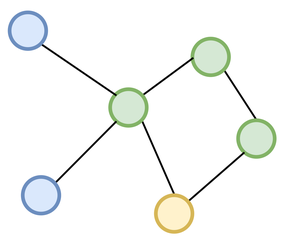
Mentions, Relations, Source text

Normalization and Disambiguation



Aliases and Instances → Canonical entities

MAGKG



Aggregated nodes and relations

Application — Sec.5.5

Neo4j Graph Store

Cross-Deposit Associations

Path and Evidence Analysis

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